

CHECKLIST

SMALL CRAFT - LIQUEFIED PETROLEUM GAS (LPG) SYSTEMS

Ref.: EN ISO 10239:2017 (ISO 10239:2014)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



**Note: The scope does not cover devices used for LPG-fuelled propulsion engines or
LPG-driven generators.**

Subject to check	Clause	Requirements	Checked ?
1 The system is of a vapour withdrawal type, i.e. LPG released only under gas phase conditions.	4.2	[Yes ?]	
2 All appliances installed on a single LPG system shall be designed for use at the same operating pressure and the same LPG type, e.g. propane, butane, or a mixture of the two.	4.3	[Yes ?]	
3 Label: Operating pressure. Affixed: In the vicinity of the cylinder shut-off valve.	4.3	[Yes ?]	
4 The cylinder(s) selected and other supply equipment has sufficient capacity to ensure safe and satisfactory operation of all appliances simultaneously. Cylinder locker or cylinder housing is capable of accommodating the capacity of cylinders needed.	4.3	[Yes ?]	
5 Where an additional LPG system is installed there is no connection between each of the LPG supplies. The cylinder(s) for each gas supply may be installed in the same cylinder locker or cylinder housing.	4.4	[Yes / NA ?]	
6 Label: Warning sign that there is an additional LPG supply, if applicable. Affixed: In each cylinder locker or cylinder housing.	4.4	[Yes / NA ?]	
7 Label: Indication which appliance are supplied by each LPG supply, if applicable. Affixed: Inside the cylinder locker or cylinder housing.	4.4	[Yes / NA ?]	
8 Each LPG system is fitted with simple means to test the LPG system for leakage before use of any appliances (e.g. a pressure gauge, bubble leak tester).	4.5	[Yes ?]	
9 Where a bubble leak detector is fitted in the LPG system, it shall be securely mounted in the low pressure side of the LPG system and in the cylinder housing or cylinder locker.	4.5	[Yes / NA ?]	
10 If pressure gauges are used, they shall read the cylinder pressure side of the pressure regulation device. The gauge scale shall have a pressure range from 0 kPa to a maximum of between 1000 kPa and 1600 kPa to be able to show pressure drops during the LPG system check.	4.5	[Yes / NA ?]	
11 System is equipped with, or have provision for the installation of a pressure regulation system.	5.1	[Yes ?]	

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12	Label: Working pressure of the LPG appliances Affixed: In the vicinity of the LPG cylinder installation	5.1	[Yes ?]	
13	The LPG system has a provision for an installation of a pressure regulation device or the installed LPG pressure reduction system has an overpressure device to prevent uncontrolled pressure increase in the low pressure side to a value above 5 kPa.	5.2	[Yes ?]	
14	Any gas discharge of the LPG pressure reduction system is inside the cylinder locker or housing or separately vented outside the craft.	5.2	[Yes ?]	
15	The pressure regulator indicates the nominal working pressure.	5.3	[Yes ?]	
16	The pressure regulator is not of the external manual adjustment type.	5.4	[Yes ?]	
17	The pressure regulator is located within the cylinder locker or housing.	5.5	[Yes ?]	
18	Pressure regulator, not supported by the cylinder connection, is separately secured within the cylinder locker or cylinder housing for protection, dirt and water. It is mounted above the cylinder valve for a continuous rise from the cylinder valve to the regulator.	5.6	[Yes / NA ?]	
19	The LPG supply line system is either a solid piping system (except for short hose connection to gimbaled stoves) or continuous hose in accordance with 6.3.	6.1.1	[Yes ?]	
20	Hoses are used to connect gimbaled stove(s) with their LPG supply.	6.1.2	[Yes / NA ?]	
21	Hoses are used to connect supply piping to the pressure regulation device within the cylinder locker or cylinder housing.	6.1.2	[Yes ?]	
22	If piping installed, only solid drawn copper or drawn stainless steel piping is used. Materials are galvanically compatible when connected.	6.2.1	[Yes / NA ?]	
23	There shall be no joints or fittings in piping passing through engine compartments.	6.2.2	[Yes / NA ?]	
24	LPG supply piping routed through engine compartments shall be protected by conduit or trunking, or supported by non-abrasive attachments which are no more than 300 mm apart.	6.2.3	[Yes / NA ?]	
25	Piping is installed as high as practical above bilge water level.	6.2.5	[Yes / NA ?]	
26	Piping has as few fittings as practical. Joints and fittings are readily accessible.	6.2.6	[Yes / NA ?]	
27	Hoses are routed through the engine compartment.	6.3.2	[No ?]	
28	Hoses have a minimum practical length.	6.3.2	[Yes ?]	
29	Hoses shall have permanently attached end fittings.	6.3.3	[Yes ?]	
30	Hoses are capable of being reached for inspection, removal or maintenance without removal of a permanent craft structure.	6.3.3	[Yes ?]	
31	Hoses shall be installed so as to avoid stress or tight radius turns.	6.3.3	[Yes ?]	
32	Hose connections shall be readily accessible and stress free, i.e. not subjected to tension or kinking under any conditions of use.	6.3.4	[Yes ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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33	Hoses used for LPG supply line are continuous from within the cylinder locker or cylinder housing to the appliances, or the readily accessible shut off valve near the appliance (see 6.6.3), except where metallic supply piping is connected to flexible hose leading to a movable appliance, such as a gimbaled stove.	6.3.5	[Yes / NA ?]	
34	Piping does not have direct contact with metallic parts of craft structure of higher galvanic nobility than the piping.	6.5.1	[Yes / NA ?]	
35	Supply lines and components are routed 30 mm away from electrical conductors unless the LPG line is run jointless through a conduit, or the conductors are sheathed in conduit or trunking according to ISO 10133 and ISO 13297.	6.5.2	[Yes ?]	
36	Metallic supply lines are at least 100 mm from engine exhaust system.	6.5.2	[Yes / NA ?]	
37	LPG lines are at least 100 mm from exposed electrical terminals.	6.5.3	[Yes / NA ?]	
38	Support of LPG supply lines in order to prevent damage from chafing or vibration: - copper or stainless steel piping, spaced at intervals not exceeding 0,5 m; - for hoses, the intervals shall not exceed 1 m.	6.5.4	[Yes ?]	
39	LPG supply line fixing devices are corrosion-resistant, non-abrasive, designed to prevent cutting or other damage to the lines and galvanically compatible with the supply line material. In the case of conduit it shall be vented and non-metallic. All joints shall have at least one fixing device per line no more than 150 mm away from the joint.	6.5.4	[Yes ?]	
40	No undue stress is created at the fittings at joints and connections.	6.5.5	[Yes ?]	
41	Penetrations through watertight bulkheads maintain watertight integrity.	6.5.6	[Yes / NA ?]	
42	Line is protected from abrasion at through-bulkhead or wall penetrations.	6.5.7	[Yes / NA ?]	
43	A readily accessible manually operated main shut off valve is provided in the high pressure side. This may be the cylinder valve.	6.6.1	[Yes ?]	
44	Dual cylinder system is provided with an automatic or manual change over device (selector valve), with non-return valves fitted, in addition to each cylinder shut-off valve, to prevent the escape of gas when either cylinder is disconnected.	6.6.2	[Yes / NA ?]	
45	A shut-off valve is installed in the low pressure supply line to each appliance. This may be a solenoid valve located within the cylinder locker or cylinder housing on the high or low pressure side of the pressure regulation device, operable from the vicinity of the appliance. Solenoid valves shall be closed in cases of lack of tension, i.e. loss of electrical actuating energy.	6.6.3	[Yes ?]	
46	Each shut-off valve is in vicinity of the appliance and operable without reaching over the top of open flames.	6.6.3	[Yes ?]	
47	Identification of open/closed position of shut-off valves are clearly identified.	6.6.4	[Yes ?]	

compliant: Yes or V

not applicable: NA

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48	Label: Identification of controlled appliance at shut-off valve that is placed away from the appliance. Affixed: At shut-off valve.	6.6.5	[Yes / NA ?]	
49	Label: Location of a shut-off valve for an appliance if directly visible. Affixed: At the appliance.	6.6.5	[Yes / NA ?]	
50	Taper plug valves are only used in low pressure side and are spring loaded.	6.6.6	[Yes / NA ?]	
51	Shut off valves are located to avoid inadvertent or accidental operation.	6.6.7	[Yes ?]	
52	All unattended appliances are of the room sealant type with air intake ducting and flues for outgoing combustion products to outside the craft.	7.4	[Yes / NA ?]	
53	All unattended appliances shall have a combustion system in which either: - incoming combustion air passes through sealed ductwork connected to the enclosed combustion chamber and terminating outside the craft, including any areas that can be enclosed by canopies, or - mechanisms are incorporated in the appliance to prevent back drafting from the exhaust and oxygen depletion in interior spaces.	7.4	[Yes / NA ?]	
54	Label: Type of LPG (e.g. "Butan", "Propan") and notice "see owner's manual". Affixed: At the appliance	7.5	[Yes ?]	
55	Warning label: Permanent, legible "DANGER — Avoid asphyxiation. Provide ventilation when the cooking appliance is in use. Do not use for space heating". Affixed: at in a conspicuous position on or adjacent to the cooking appliance. Size: Character height 4 mm Language: Acceptable in the country of intended use.	7.6	[Yes / NA ?]	
56	The exposed hot working surfaces of space heaters and water heaters are located to prevent risk of injury.	7.8	[Yes / NA ?]	
57	Manufacturers instructions are provided preventing overheating of surfaces and allowing inspection/servicing.	7.9	[Yes ?]	
58	For monohull sailing craft: Sliding of cooking utensils across the stove is prevented up to 15° pitch and 30° roll.	7.10	[Yes / NA ?]	
59	For monohull engine driven and multihull sailing craft: Sliding of cooking utensils across the stove is prevented for 15° pitch and roll.	7.10	[Yes / NA ?]	
60	Cylinders, regulators and safety devices are secured for marine environment.	8.1	[Yes ?]	
61	Cylinders, regulators and safety devices are installed in lockers or housings.	8.1	[Yes ?]	
62	Cylinders, pressure regulators, regulation devices and safety devices located below decks or in cockpits shall be mounted in cylinder lockers.	8.1	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

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63	Craft design and openings of cylinder lockers and cylinder housings shall be such that escaping vapours can only flow to the outside of the craft.	8.2	[Yes ?]	
64	Cylinder lockers inside enclosed cockpits are only be accessible from the top. In case of a cockpit with open transoms it may also be accessible from the side.	8.3	[Yes / NA ?]	
65	A cylinder locker shall be vented at the bottom by a drain with not less than 19 mm internal diameter; flanges or welded joints 30 mm above the lowest point of the locker are accepted.	8.3	[Yes / NA ?]	
66	The locker drain runs outboard without sumps which can retain water.	8.3	[Yes / NA ?]	
67	The locker drain runs outboard with outlet lower than locker bottom, not less than 75 mm above waterline at fully loaded condition.	8.3	[Yes / NA ?]	
68	All hoses and metal piping penetrating the locker wall are sealed vapour tight.	8.4	[Yes / NA ?]	
69	Locker drains and housing vents are located at least 500 mm away from any opening to the interior of the craft.	8.5	[Yes / NA ?]	
70	No store for loose storage or components is provided in the locker.	8.6	[Yes ?]	
71	Cylinders, valves and pressure regulators are readily accessible and secured rigidly allowing only withdrawal of gas in vapour condition.	8.7	[Yes ?]	
72	All electrical devices in cylinder lockers, housings or compartments comply with ISO 8846 for ignition protection.	11	[Yes / NA ?]	
73	Flues are routed and sized to ensure complete discharge outside craft, including areas that maybe enclosed by canopies and as not to be obstructed by water.	13.2	[Yes / NA ?]	
74	The flue and air-intake duct system is continuous and vapour tight from the appliance to its terminal outside the craft.	13.3	[Yes / NA ?]	
75	Dampers (shut-off valves) are not used in the flue system.	13.4	[Yes / NA ?]	
76	The entire flue system is accessible for inspection.	13.5	[Yes / NA ?]	
77	Flue terminals for exhaust discharge are not within 500 mm of a ventilator, opening port, window, refuelling fitting or fuel tank vent.	13.6	[Yes / NA ?]	
78	Flue terminals are constructed with guard to prevent damage and injury by accidental contact with hot surfaces.	13.7	[Yes / NA ?]	
79	If cooking appliances with integral LPG cartridges, capacity of 225 g or less.	Annex D	[Yes / NA ?]	
80	Cartridge has a self-closing device to enable its removal for storage when not in use.	D.2	[Yes / NA ?]	
81	Operating controls shall be readily accessible, and located to minimize possible injury from burners or elements when being used.	D.4	[Yes / NA ?]	
82	Burner controls shall be equipped or designed to require two-stage operation when going from the "off" to "on" position to prevent unintentional or accidental opening of valves during handling and storage.		[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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83	Cooking appliances in use shall have a positive means of mechanical retention and be secured in a location designated by the boat manufacturer.	D.7	[Yes / NA ?]	
84	Means shall be provided on or adjacent to stove top cooking surfaces to prevent both deep and shallow cooking pans from sliding across or off the stove during craft motion; see also 7.9.	D.7	[Yes / NA ?]	
85	Alternatively, guidance shall be provided to use the cooking appliance only when safe to do so.	D.7	[Yes / NA ?]	
86	Reserve or empty cartridges are not stored inside the boat but only on the boat exterior, protected from the weather and mechanical damage, and where escaping vapours can only flow towards the outside.	D.8	[Yes / NA ?]	
87	Printed instructions are provided with each cooking appliance.	D.9	[Yes / NA ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

Subject to check	Clause	Requirements	Checked?
88 The system and all components withstand storage from -30 °C to +60 °C.	4.1	[Yes ?]	
89 Regulating system is designed to provide a fixed nominal pressure suitable for the consuming appliances, but not more than 5 kPa.	5.1	[Yes ?]	
90 LPG pressure reduction system is a pressure relief governor, a pressure relief valve or an automatic safety shut off valve.	5.2	[Yes ?]	
91 The pressure regulator and its fastener are made of corrosion-resistant metallic material or have an effective coating against external corrosion.	5.7	[Yes ?]	
92 The piping and hose are sized not to drop working pressure below required operating pressure at any appliance below that required by the appliance manufacturer when all appliances are operating simultaneously.	6.1.3	[Yes ?]	
93 The minimum wall thickness for piping with outside diameter ≤ 12 mm is 0,6 mm, and 0,9 mm for diameters > 12 mm.	6.2.1	[Yes / NA ?]	
94 Semi-rigid, pliable corrugated stainless steel tubing (PCT) shall conform to EN 15266, or equivalent.	6.2.1	[Yes / NA ?]	
95 Fittings for connections and joints in piping shall be metallic and of a proper type in accordance with the standard.	6.2.4	[Yes ?]	
96 Jointing compound for flared fittings or flared rings and gas tightness by compression of ductile joints (except connections in accordance with EN 16129:2013, Annex M) shall not be used.	6.2.4	[Yes / NA ?]	
97 Materials and components of hose assemblies are designed to be suitable for LPG and to withstand the stresses and exposures found in the marine environment.	6.3.1	[Yes ?]	
98 Permanently attached end fittings of hoses are swaged sleeve or sleeve and threaded insert.	6.3.3	[Yes ?]	
99 The melting point of materials at welded or brazed connections is below 450 °C.	6.4.1	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

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100	Fittings through which LPG passes are compatible with LPG and galvanically compatible with the metallic piping to which they are connected.	6.4.2	[Yes ?]	
101	Hose clamps, if used to secure cylinder locker vent hoses, are corrosion resistant and reusable.	6.4.3	[Yes / NA ?]	
102	End connection fittings are corrosion resistant.	6.4.4	[Yes ?]	
103	Where cutting ring fittings are used in conjunction with copper piping, a brass insertion sleeve and brass cutting ring shall be fitted. All components match to avoid galvanic corrosion.	6.4.5	[Yes / NA ?]	
104	Threaded gas tight connections are of the taper pipe thread type conforming to ISO 7-1, or fittings conforming to EN 1949.	6.5.8	[Yes / NA ?]	
105	For threaded gas tight connections sealants are used conforming to EN 751-2 or EN 751-3.	6.5.8	[Yes / NA ?]	
106	Only appliances for use in marine environment are used in the LPG system.	7.1	[Yes ?]	
107	The appliances are fitted in accordance with the manufacturers instructions.	7.1	[Yes ?]	
108	Each appliance is securely fixed as to eliminate undue stress to piping, hoses and fittings.	7.2	[Yes ?]	
109	Each appliance has a flame supervision devices for each burner and/or pilot lights.	7.3	[Yes / NA ?]	
110	Needle valves are used as shut-off valves in low pressure side of system and gate valves are not used as shut-off valves.	6.6.8	[Yes ?]	
111	If the incoming air is not delivered through sealed ductwork terminating outside the craft, and if the appliance is installed in interior spaces, ventilation shall be provided that allows outside air to pass through fixed openings (Annex B).	7.4	[Yes / NA ?]	
112	Ventilation is provided in accommodation spaces where open flame unflued appliances are used or to which compartments containing such appliances are connected by open passageways. Minimum sizing and locations of ventilation openings complies with Annex B.	9	[Yes / NA ?]	
113	Prior to charging the system with LPG, the supply line and fittings have been tested with air; test pressure three times the nominal pressure but not more than 15 kPa.	10.	[Yes ?]	
114	No pressure drop was indicated after a period of 10 min; in case that any leakage has been indicated by a drop in pressure, the entire LPG system has been checked.	10.	[Yes ?]	
115	If the pressure regulating device is not rigidly connected to, and supported by, the cylinder connection, high pressure side components are checked for leakage.	10.	[Yes / NA ?]	
116	Flue components are installed with the manufacturer's instructions.	13.1	[Yes / NA ?]	
117	The minimum effective area of ventilation is given.	Annex B	[Yes ?]	
118	Cooking appliances is suitable for use with LPG in a marine environment and installed in accordance with the manufacturer's instructions.	D.1	[Yes / NA ?]	
119	ISO 9094 is meet regarding the proximity and flammability of materials.	D.1	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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120	The cooking appliance is complies with the design specifications.	D.3.	[Yes / NA ?]	
121	Appliance has a continuously burning pilot light.	D.6	[Yes / NA ?]	

Instructions/Warnings to be included in the owner's manual				
122	WARNING — Never leave craft unattended when open flame LPG consuming appliances are in use.	Annex C	[Yes ?]	
123	WARNING — Do not smoke or use open flame when replacing LPG cylinders. Close cylinders before disconnecting for replacement.ylinder valves on empty cylinders.	Annex C	[Yes ?]	
124	WARNING — Never use a flame to check for leaks.	Annex C	[Yes ?]	
125	WARNING — Fuel burning open flame appliances consume cabin oxygen and release products of combustion into the craft; Do not use the stove or oven for space heating. Ventilation is required when appliances are in use. Open designated vent and openings while appliances are in use. Never obstruct ventilation openings. The ventilation requirements have been calculated to suit the LPG appliances as installed. Additional ventilation might be required if other appliances are operated simultaneously.	Annex C	[Yes ?]	
126	CAUTION — Do not use solutions containing ammonia for manual leak testing.	Annex C	[Yes ?]	
127	Warning label: DANGER — Avoid asphyxiation. Provide ventilation when the cooking appliance is in use. Do not use for space heating.	Annex C	[Yes / NA ?]	
128	WARNING — Do not modify the craft's LPG system. Installation, alterations and maintenance shall be performed by a competent person. Have the system inspected at regular intervals or as required by national requirements.	Annex C	[Yes ?]	
129	WARNING — If a leak is detected shut off the main LPG supply valve and do not use LPG appliances.	Annex C	[Yes ?]	

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130	General	Annex C	[Yes ?]	
	— Operating pressure of the LPG system; — Location and type of ventilation openings (e.g. closable ventilators such as windows, hatches, mushroom and dorade ventilators) in accommodation spaces containing LPG appliances; — Recommendation not to obstruct access to LPG system components in any way; — If a second LPG system is fitted, the owner's manual shall include information on which appliances are connected to each LPG system and the operating pressure of each LPG system; — Regular inspections of hoses and flue pipes in the LPG system, at least annually, and replacement if any deterioration is found; — Valves on empty cylinders shall be kept closed and disconnected. Protective covers, caps or plugs shall be kept in place. Reserve or empty cylinders shall be stored in LPG cylinder lockers or housings which are vented to the outside and intended for that purpose or on the boat exterior, protected from the weather and mechanical damage, and where escaping vapours can only flow overboard; — LPG cylinder housings or cylinder lockers shall not be used for storage of any other equipment.			
131	Operation of the LPG system	Annex C	[Yes ?]	
	— LPG supply line valves and cylinder valves shall be closed when appliances are not in use, before refuelling and immediately in an emergency. — Appliance valves shall be closed before opening cylinder valve. — The user shall be informed on the need for ventilation when operating appliances that consume cabin oxygen. This information shall also be provided by the safety notice, see C.4. — If the stove is not gimballed, it shall not be used when high angles of rolling or sustained angles of heel are likely.			
132	The owner's manual instructs the operator to remove and to replace cartridges in the open air and away from sources of ignition.	D.2	[Yes ?]	

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133 LPG system check

Annex C

[Yes ?]

The LPG system shall be checked for leakage before each use by means of the following:

a) ~~With~~ with a gauge fitted: close appliance valve, open LPG cylinder valve, allow indicated gauge pressure to stabilize, close LPG cylinder valve, observe pressure gauge reading for 3 min, pressure gauge reading should remain constant if no leak in the LPG system is present, or

The manual shall specify that the gauge does not provide an indication of liquid LPG remaining in the cylinder, only its vapour pressure, which is a constant at any given temperature.

b) with a bubble leak detector fitted: operate as per manufacturer's instructions.

If LPG leakage is detected or suspected, the following action shall be taken immediately:

- a) Shut off the LPG supply at the main supply valve(s);
- b) Extinguish naked flames and other ignition sources (heaters, cooking appliances, pilot lights etc.);
- c) Do not operate electric switches;
- d) Evacuate the area if possible.

Warning — Do not use an installation that has leaked until it has been inspected and repaired by a competent person.

The owner's manual shall include a clarification that the above user tests do not replace a LPG system check by a competent person and a warning not to use a flame to check for leaks. See example safety notices as provided in C.4.

Comments:
